

Jianliang He

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RESEARCH INTERESTS

Machine Learning Theory; Mechanistic Interpretability; Large Language Model.

EDUCATION

Yale University

Department of Statistics and Data Science

Ph.D. in Statistics.

Advisor: Prof. Zhuoran Yang.

2024.9 - Present

New Haven, CT

Fudan University

Department of Statistics and Data Science, School of Management

B.S. in Statistics

2020.9 - 2024.6

Shanghai, China

RESEARCH PAPERS

* stands for equal contribution or alphabetical ordering.

1. **He, J.**, Wang, L., Chen, S., Zhang, F., Yang, Z. "Neural Networks Provably Learn Spectral Representations for Group Composition". [arXiv:2606.02993](https://arxiv.org/abs/2606.02993). Submitted, 2026.
2. **He, J.**, Wang, L., Chen, S., Yang, Z. "On the Mechanism and Dynamics of Modular Addition: Fourier Features, Lottery Ticket, and Grokking". [arXiv:2602.16849](https://arxiv.org/abs/2602.16849). Submitted, 2026.
3. Wei, J., Chen, S., **He, J.**, Yang, Z. "How Transformers Learn Causal Structures In-Context: Explainable Mechanism Meets Theoretical Guarantee". *International Conference on Learning Representations (ICLR)*, 2026.
4. **He, J.**, Pan, X., Chen, S., Yang, Z. "In-Context Linear Regression Demystified: Training Dynamics and Mechanistic Interpretability of Multi-Head Softmax Attention". [arXiv:2503.12734](https://arxiv.org/abs/2503.12734). *International Conference on Machine Learning (ICML)*, 2025.
5. Qin, S.*, **He, J.***, Kuang, Q*, Gang, B., Xia, Y. "Data-light Uncertainty Set Merging with Admissibility". [arXiv:2410.12201](https://arxiv.org/abs/2410.12201). Submitted, 2024.
6. **He, J.***, Chen, S*, Zhang, F., Yang, Z. "From Words to Actions: Unveiling the Theoretical Underpinnings of LLM-Driven Autonomous Systems". [arXiv:2405.19883](https://arxiv.org/abs/2405.19883). *International Conference on Machine Learning (ICML)*, 2024.
7. **He, J.**, Zhong, H., Yang, Z. "Sample-Efficient Learning of Infinite-Horizon Average-Reward MDPs with General Function Approximation". [arXiv:2404.12648](https://arxiv.org/abs/2404.12648). *International Conference on Learning Representations (ICLR)*, 2024.
8. Banerjee, T*, Gang, B*, **He, J.***. "Harnessing the Collective Wisdom: Fusion Learning using Decision Sequences from Diverse Sources". [arXiv:2308.11026](https://arxiv.org/abs/2308.11026). *Biometrika*, 2026.

INDUSTRIAL EXPERIENCE

Machine Learning Engineer, Cisco Foundation AI Team, San Francisco

2025.6 - Present

- Led the development of cybersecurity-focused reasoning Large Language Models with a technical report available at [arXiv:2601.21051](https://arxiv.org/abs/2601.21051), and worked on Agentic Reinforcement Learning.

TEACHING

Teaching Assistant, Yale University

- S&DS 241 Probability Theory
- S&DS 265 Introductory Machine Learning

Fall, 2025

Spring, 2026

Teaching Assistant, Fudan University

- MANA130083.01 Nonparametric Statistics

Spring, 2023

SERVICE

Conference Reviewer: NeurIPS, ICLR, ICML.

Journal Reviewer: Management Science.